

DeltOS — Complete Project Handoff

Goals, architecture, all 87 components, technology selections, format specification, normative values, implementation state and open decisions

September 2026 · self-contained · intended as the sole input to an implementing system

0. How To Use This Document

This is a complete handoff, written to be the only document an implementing system needs. It carries the project's goals, the full component inventory, every technology selection, the container format's byte layout, the normative values shared between components, the current implementation state with measured results, and the decisions still open.

- **Sections 1–3, context:** What the system is for, what constrains it, how the work divides.
- **Sections 4–9, specification:** Complete component inventory, technology chosen per capability, the WAX container format, and normative cross-component values. Implementers work from these.
- **Sections 10–13, state:** What exists and is measured, what is blocked pending a decision, known defects, and phase ordering.
- **Section 14, method:** How implementation has been run against these documents, and the rule that kept it correct.

Where a value is stated as normative it is authoritative and must not be re-derived. Where a decision is stated as open it must not be resolved silently — an implementer needing an open value should surface it rather than choose one. Five implementation passes have halted for exactly this reason, and every halt was correct: each had found a real contradiction, a stale document, or a value three other components were about to depend on.

1. What DeltOS Is

DeltOS is an offline-first education and library appliance: a small computer — as small as a Raspberry Pi Zero 2 W — serving a library of content and a suite of services to everyone on its local network, with no internet connection required at any point, ever.

Target deployments are places where connectivity is absent, unaffordable or unreliable: a one-room school, a rural clinic, a public library, a disaster-response site, an off-grid research station. In each, the box is the internet. It holds Wikipedia, a learning platform, media, documents, chat and whatever else the deployment needs, and serves it to phones and laptops that join its own Wi-Fi.

What distinguishes it

- **A storage format built for incremental update:** DeltOS defines the WAX container. The dominant existing format in this space is write-once, so updating content means re-downloading the whole archive. WAX is a segment chain: an update transfers only what changed. Over a satellite link or a sneakernet USB drive that is the difference between a library that stays current and one frozen at its install date.
- **An operating layer, not a content reader:** The learning platform, media server, office suite, wiki, chat and mail install and launch the same way a content pack does — behind one login, on one box, from one admin console. Making forty services feel like one appliance is the product.
- **Designed down to 512 MB, not up from a workstation:** The lowest supported tier is a Pi Zero 2 W. Capability degrades by measured hardware resources; a design that only works on generous hardware has failed the deployments that need this most.

The name is from δέλτος (deltos), the ancient Greek wax writing tablet, pairing with the WAX format — the wax the tablet holds. The shell layer was called "Atrium" in early drafts; that name is retired.

2. Goals and Design Principles

These are the constraints that decide arguments. Where a proposal conflicts with one of these, the principle wins unless it is explicitly revisited.

- **Offline is the default, never a degraded mode:** Nothing assumes a connection to render, search, translate, narrate or update. Connectivity, where present, is an optimization.
- **The floor is a 512 MB box:** Features degrade by measured hardware capability rather than being absent or broken. Memory and storage budgets are stated and tested at the floor, not at the ceiling.
- **Incremental updates over whole re-downloads:** Applies to content packs, the catalog, and OS images alike.
- **Reuse over rebuild:** Where a mature open-source component fits, DeltOS integrates it rather than reimplementing it. What DeltOS builds is the layer that unifies them.
- **Prefer single-binary, Rust or Go, SQLite- or flat-file-backed:** A consistent operational model: small footprint, clean ARM cross-compilation, no heavy runtime. Heavier stacks (Nextcloud, Kolibri) are admitted deliberately and named, never by default.
- **A crashed service repairs itself:** Field deployments have no administrator on call. Self-healing orchestration is baseline, not refinement.
- **Nothing ships unexamined:** Licensing is validated at build time, packs are signed, and an unverifiable pack does not install.
- **Refuse rather than guess:** Readers reject unknown major versions, unknown compression, and malformed input rather than attempting recovery. The same posture applies to implementers: an undefined value is escalated, not invented.

3. Architecture

3.1 The eight tracks

Track	Owns	Discipline
A	The .wax container format, reader/writer library, builder, signing, streaming server, delta engine.	Systems / Rust
B	Content and interop: ZIM conversion, crawling, manifests, on-device catalog, publishing and review pipeline.	Content pipeline
C	The DeltOS Shell: kiosk launcher, unified search, profiles, sandboxing, onboarding, accessibility.	Frontend / UX
D	Search and intelligence: full-text and semantic search, tiered local LLM, retrieval-augmented answering.	ML / search
E	Hardware, OS image, atomic updates, Wi-Fi and captive portal, mesh, IoT and radio.	Systems / infra
F	Admin console, identity and roles, install pipeline, device health, fleet, governance.	Product / security
G	Platform services: orchestration, reverse proxy, git, registry, package cache, automation, metrics.	Platform / DevOps
H	The application catalog: LMS, media, office, wiki, chat, VoIP, mail, storage, browser IDE.	Full-stack / apps

3.2 Hardware tiers — normative

Guaranteed floors, not ranges. The floor is what any component may assume; headroom above it is discovered at runtime and never inferred from the tier name.

Tier	Min RAM	Min storage	Arch	Notes
pi_zero_2w	512 MB	16 GB	aaarch64	Kiosk floor. No platform services, no applications.
pi_4	2 GB	32 GB	aaarch64	A 1 GB Pi 4 SKU exists and is below this floor — it is not this tier.
pi_5	4 GB	64 GB	aaarch64	First tier meeting the local-LLM floor.
mini_pc	16 GB	256 GB	x86_64	Community Hub and Field Ops floor.
generic	declared	declared	either	Installable-stack hosts only. Box-reported; never declared by a pack.

- **Total ordering:** pi_zero_2w < pi_4 < pi_5 < mini_pc. A pack or feature declaring a minimum tier runs on that tier and every tier above it.
- **Tier gates cataloging; measured resources gate runtime:** A tier is a floor, so a pi_5 box may have 4 GB or 16 GB. A feature needing 8 GB must ask what the box actually has. Every box publishes both its tier and its measured RAM, storage and architecture.

3.3 Deployment profiles — normative

A profile states what a deployment turns on. It is a separate axis from the hardware tier, which states what the hardware is. Conflating the two is a recurring error and is explicitly forbidden.

Profile	Wire value	Min tier	What it adds
Kiosk	kiosk	pi_zero_2w	Core format, content, search, plus baseline identity daemons. No platform services, no applications.
Classroom	classroom	pi_4	Full shell; baseline orchestration and ingress; standard applications; captive portal.
Community Hub	community_hub	mini_pc	Full platform services; most of the application catalog.
Field Ops	field_ops	mini_pc	Community Hub plus opt-in IoT, SDR, RF and smart-grid modules.

- **Total ordering:** kiosk < classroom < community_hub < field_ops. "Classroom and above" resolves against this and nothing else.
- **Profile is chosen; tier is detected:** A profile is selected at provisioning or pre-seeded onto an image. A tier is measured at first boot. A profile may not be selected on hardware below its minimum tier.

3.4 Two co-equal hardware paths

- **Appliance image:** An rpm-ostree-based immutable image with A/B atomic updates and rollback. Recommended for hardware dedicated to DeltOS; gives the strongest update guarantee.
- **Installable stack:** The containerized stack laid onto an existing general-purpose Debian/Ubuntu-family host, including a VPS or repurposed desktop. Closer to how comparable systems are deployed today.

Both are available on every profile. A profile says what runs, not which path put it there. Installable-stack hosts report the generic tier with measured resources.

4. Complete Component Inventory — 87 Components

Every numbered component across the eight tracks. The phase column gives the target phase (P0 earliest); "built" marks what exists today. Phase ordering is a dependency ordering, not a schedule of dates.

Track A — Format & Storage Engine (10)

ID	Name	Description	Phase
A1	wax-core	Rust reader/writer library for the .wax container. Both sides stream; memory does not scale with archive size.	built
A2	wax-builder	CLI assembling a directory tree into a signed, compressed .wax archive. Full manifest enforcement, append, build report.	built
A3	SPEC.md	The formal, versioned byte-layout specification. Frozen; amended only via documented precedence.	built
A4	Conformance suite & fuzzers	Reference test corpus plus cargo-fuzz targets for the reader. 205 tests, three fuzz targets.	built
A5	wax-serve	HTTP server exposing byte-range and streaming reads. The shell cannot open a pack without it.	P1
A6	wax-delta	Append/patch engine computing deltas between pack versions using content-defined chunking (FastCDC).	P4
A7	Pack signing	Detached minisign signatures now; TUF-based multi-key signing later. Build-time half is built; enforcement lands P2.	P2
A8	Multi-volume archives	Splitting one library across multiple physical volume files; the index is never split, only blob data.	P3
A9	Embedded search-index section	SQLite FTS5 table inside the index footer at v0.9; a stand-off Tantivy segment at v2.	P2
A10	Reference reader (2nd language)	A reader in a second language, proving the spec is portable rather than merely implemented once.	P1

Track B — Content & Interop (13)

ID	Name	Description	Phase
B1	zim2wax	Converts existing ZIM archives into WAX packs. Working against real Wikipedia.	built
B2	Crawler pipeline	Site-to-pack tooling for original content, reusing Browsertrix Crawler plus a warc2wax converter.	P2
B3	Pack manifest schema	Name, icon, category, license, attribution, version, hardware floor, entry point, runtime costs, dependencies.	built
B4	On-device catalog	Local SQLite database of installed and available packs, synced from the publishing service.	P1
B5	Starter packs	Curated grade- and subject-tagged bundles shipped for first run.	P2
B6	Licensing/attribution validator	Build-time SPDX allowlist check with three outcomes: clean, review-required, or hard failure.	built
B7	WAX Studio	Curation and authoring tool, separate from the runtime.	P5
B8	Incremental re-crawl tooling	Updates a pack from its source without a full re-crawl, preserving pack identity.	P5
B9	Community submission & review	Intake and moderation pipeline for contributed packs: automated checks, human review, sign, publish.	P2
B10	Offline mapping pipeline	OSM extracts, OSRM routing graphs and offline tiles packaged as a standard WAX pack.	P3
B11	Offline geocoding	Forward geocoding via full-text search over place names; SQLite R-Tree for reverse geocoding only.	P3
B12	DevDocs offline pack	Mirrors devdocs.io documentation sets as a content pack rather than a	P2

		running service.	
B13	Catalog publishing service	Central infrastructure the review pipeline feeds; the sole point where the trusted project key signs.	P2

Track C — DeltOS Shell (11)

ID	Name	Description	Phase
C1	DeltOS Shell	Full-screen kiosk launcher, split into an unprivileged web app and a privileged native helper over a narrow IPC surface.	P0/P1
C2	Pack/app registry	Populates the launcher grid from the on-device catalog; multi-sourced across content packs and services.	P1
C3	Unified search bar	One query across every installed pack, through the search service's single endpoint.	P2
C4	Per-pack sandbox	iframe plus CSP isolation now; WASM (Wasmtime) isolation later. Per-pack, per-profile origins.	P1
C5	Multi-profile / classroom mode	Separate students and classes on one shared device, with storage isolation per profile slot.	P2
C6	Offline app store UI	Browsing the on-device catalog fully offline, including a hardware requirement calculator.	P2
C7	USB / peer install flow	Installing from a USB drive or a peer box exactly like installing from a catalog.	P2
C8	Coach/progress dashboard	Per-learner progress for teachers, reading the shared progress event store.	P3
C9	Accessibility layer	Text-to-speech (Piper), translation, high-contrast, large text, reader mode, screen magnifier.	P2
C10	First-boot onboarding wizard	Guided setup with no raw config files exposed; language and accessibility chosen before network.	P1
C11	Ask DeltOS	Conversational interface built into the shell as a first-class app, calling the RAG engine.	P4

Track D — Search & Intelligence (8)

ID	Name	Description	Phase
D1	Full-text search service	Embedded SQLite FTS5 per pack at v0.9–v1.x, moving to Tantivy at v2; an aggregation service gives one query endpoint.	P1
D2	Vector/semantic index	sqlite-vec or LanceDB alongside the text index.	P4
D3	Multilingual tokenization	CJK, Indic and Swahili-family coverage at launch, selected by the pack's declared languages.	P1
D4	Build-time embedding pipeline	Content embedded once at pack-build time, never on the device.	P4
D5	Local LLM runtime	llama.cpp with GGUF models, tiered by hardware: no model at Kiosk, a small model at pi_5, a larger one at mini_pc.	P3
D6	RAG orchestration	Retrieves passages and answers with citations back to pack and entry path.	P4
D7	Hardware-tier feature gating	One codebase, graceful degradation; the single source of truth for every tier decision.	P2
D8	Federated search	Optional cross-box query fan-out over the mesh.	P5

Track E — Hardware, OS & Networking (15)

ID	Name	Description	Phase
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E1	Base OS image	Minimal, read-only-root image. rpm-ostree / Fedora IoT preferred; Yocto is the documented Kiosk fallback.	P0/P1
E2	A/B atomic updates	Two deployments with automatic rollback to last-known-good. Applies down to the Kiosk profile.	P1
E3	Wi-Fi AP & captive portal	Access point plus portal, rebuilt to handle documented Android/macOS/AP+STA friction correctly.	P1
E4	Network diagnostics	Plain-language connectivity troubleshooting exposed to the admin console.	P2
E5	Storage/partition layout	Read-only OS partition plus a writable data partition rooted at a single DeltOS state tree.	P0
E6	Hardware-tier profiles	The authoritative tier definitions: floors for RAM, storage and architecture.	P0
E7	Provisioning/flashing tool	Guided imaging for admins building new boxes; also the installable-stack installer.	P2
E8	Mesh networking module	Yggdrasil or Reticulum for multi-box, off-Wi-Fi topologies.	P4
E9	Power/solar dashboard	Charge state and panel-positioning guidance for the solar hardware configuration.	P4
E10	Local DNS & service discovery	CoreDNS plus Avahi/mDNS so services resolve as friendly LAN hostnames under deltos.lan.	P1
E11	Local NTP	chrony, optionally GPS-disciplined, for sites with no internet access ever.	P1
E12	IoT gateway bridge	Bluetooth, NFC and Z-Wave device bridging onto a shared MQTT bus (Mosquitto).	P4
E13	Community sensing (SDR)	Receive-only AIS, ADS-B and weather-satellite decoding via RTL-SDR. Passive reception only.	P5
E14	Advanced RF: private cellular	Open5GS/srsRAN private LTE/5G testbed. Regulatory-sensitive, opt-in, never bundled by default.	P5+
E15	Smart-grid/power automation	Modbus and MQTT integration for solar, battery and grid telemetry, driven by the automation engine.	P4

Track F — Admin, Identity & Fleet (13)

ID	Name	Description	Phase
F1	Admin console	One unified, wizard-driven web UI at a fixed local address.	P1
F2	First-boot setup	Forces a unique generated admin password, displayed once as text and QR with explicit confirmation.	P0
F3	User/role management	Closed role set: admin, teacher, student, guest. Every privileged operation is role-checked.	P1
F4	Content install/update pipeline	Resumable, queued, atomic installs with signature and rollback verification.	P2
F5	Device health monitoring	Storage, uptime, temperature and connectivity snapshots on a fixed interval.	P2
F6	Fleet dashboard	Multi-box view for organizations, with cross-site pairing over the mesh.	P4
F7	Signing & key management	Operationalizes the pack signing keys, then TUF.	P0
F8	Governance	Specification hosting, contribution guidelines, license posture, public roadmap.	P0
F9	Documentation site	Admin and deployment guides.	P2
F10	Backup/restore tooling	Content and per-learner progress data, with consistent snapshots of live databases.	P3
F11	Identity & progress data	The shared record of who a learner is and what they have done. Settled ahead	P2

	model	of its target phase.	
F12	SSO / identity provider	Authelia in front of the reverse proxy: one login across every self-hosted service.	P4
F13	Password vault	Vaultwarden for admins managing shared credentials. Admin-only scope.	P4

Track G — Platform Services & Orchestration (7)

ID	Name	Description	Phase
G1	Service orchestrator	Docker Compose with healthchecks as the single authoring format. A crashed service is detected and restarted automatically.	P2
G2	Reverse proxy / ingress	Caddy or Traefik fronting every service under one hostname with automatic internal TLS. The most load-bearing single component.	P2
G3	Local git service	Gitea or Forgejo, SQLite-backed, fed through the browser.	P4
G4	Local container registry	Lightweight OCI registry for fleet-wide image distribution with no internet.	P4
G5	Package cache	apt-cacher-ng proxy so a fleet stops re-downloading the same OS packages.	P4
G6	Visual flow-automation engine	Node-RED or equivalent, wiring hardware, APIs and services together, including physical alerts.	P4
G7	Metrics & dashboard stack	Prometheus or VictoriaMetrics plus Grafana, graphing device and community metrics over time.	P4

Track H — Applications & Community Services (10)

ID	Name	Description	Phase
H1	Learning Management System	Kolibri as the primary LMS, reaching the coach dashboard through a dedicated sync bridge.	P3
H2	Media server	Jellyfin library server for attached-storage video and music, browsed through the shell.	P2
H3	Browser office suite	Collabora Online (CODE) integrated with cloud storage.	P4
H4	Browser IDE	code-server — VS Code in the browser.	P4
H5	Cloud storage & sharing	Nextcloud, with per-profile accounts and quotas.	P3
H6	Wiki	DokuWiki (flat-file) on constrained hardware; Wiki.js on Community Hub tier.	P3
H7	Publishing / CMS	Grav (flat-file) on constrained hardware; WordPress available on Community Hub tier.	P4
H8	Secure chat & collaboration	Conduit — a lightweight Rust Matrix homeserver, preferred over Synapse for this hardware range.	P3
H9	Email server & webmail	Maddy plus Roundcube, scoped to the local network and fleet; no internet mail.	P4
H10	VoIP / voice & video	Element Call plus coturn, built on the Matrix stack rather than a separate Asterisk deployment.	P4

Flagged, not yet owned by any track

- **Download-time website configurator:** Referenced as an existing, expected feature by three track documents — a configurator a downloader interacts with before or during obtaining DeltOS, feeding the hardware-requirement calculator's selections into the actual build. It fits cleanly inside none of the eight tracks and no track has claimed it. Listed so it is not lost between them; needs an owning track before implementation prompts are drawn up.

5. Technology Selections

The selection filter: prefer single-binary, Go- or Rust-native, SQLite- or flat-file-backed tools over heavier Python, Node or JVM stacks wherever a credible option exists. Each is a fraction of the memory footprint, cross-compiles to ARM cleanly, and fits the same operational model as the format itself. Not every capability has a lightweight option — Nextcloud and Kolibri are both real, heavier stacks — and those are admitted deliberately rather than by default.

Capability	Selected	Rationale / notes
Container format	WAX (own format)	Segment-chained for incremental update; SQLite index footer.
Pack signing	minisign → TUF	Detached sidecar now; multi-key role-based signing at v2.
Compression	zstd	Per-entry; already-entropy-coded media stored verbatim.
ZIM conversion	zim 0.5.0 (pure Rust)	Exposes raw namespace bytes, which the path rules require. No libzim, no C++ dependency.
Web crawling	Browsertrix Crawler + warc2wax	Reuses maintained JS rendering, rate limiting and dedup.
Full-text search	SQLite FTS5 → Tantivy	FTS5 through v1.x; Tantivy at v2 once the stand-off segment lands.
Vector search	sqlite-vec or LanceDB	Alongside the text index.
Local LLM	llama.cpp + GGUF	Tiered by measured RAM; absent at Kiosk.
Text-to-speech	Piper	Runs locally on every tier; no cloud dependency.
Base OS	rpm-ostree / Fedora IoT	Atomic deployments with native rollback. Yocto is the documented Kiosk fallback.
Local DNS	CoreDNS + Avahi/mDNS	Serves deltas.lan; mDNS for discovery.
DHCP	Kea or systemd-networkd	Required alongside the AP; dnsmasq was rejected so this must be named explicitly.
NTP	chrony	Optionally GPS-disciplined for permanently offline sites.
Mesh	Yggdrasil or Reticulum	Multi-box topologies off Wi-Fi.
IoT bus	Mosquitto (MQTT)	Shared bus; Z-Wave JS UI and equivalents bridge onto it.
SDR	RTL-SDR: dump1090, AIS-catcher	Receive-only. Passive reception is legal near-universally.
Private cellular	Open5GS / srsRAN	Regulatory-sensitive; opt-in, never default.
Orchestration	Docker Compose	One authoring format everywhere. K3s reserved for genuine multi-box clustering only.
Reverse proxy	Caddy or Traefik	Internal TLS from a local CA; forward-auth gate in front of services.
Git hosting	Gitea / Forgejo	SQLite-backed, single binary.
Registry	Lightweight OCI registry	Fleet-wide image distribution with no internet.
Package cache	apt-cacher-ng	Fleet stops re-downloading OS packages.
Automation	Node-RED	Visual flow engine; also drives smart-grid automation.
Metrics	Prometheus/VictoriaMetrics + Grafana	Device and community metrics over time.
SSO	Authelia	Forward-auth in front of the reverse proxy.
Password vault	Vaultwarden	Bitwarden-compatible, Rust. Admin-only scope.
LMS	Kolibri	Reuse, not rebuild. A heavier stack, admitted deliberately.
Media server	Jellyfin	Attached-storage video and music.

Office suite	Collabora Online (CODE)	Lighter fit than OnlyOffice for this hardware range.
Browser IDE	code-server	VS Code in the browser.
Cloud storage	Nextcloud	A heavier stack, admitted deliberately.
Wiki	DokuWiki / Wiki.js	Flat-file on constrained hardware; database-backed above.
CMS	Grav / WordPress	Flat-file on constrained hardware; WordPress on Community Hub.
Chat	Conduit (Matrix)	Rust, lightweight; preferred over Synapse for this hardware.
VoIP	Element Call + coturn	Built on the Matrix stack rather than a separate Asterisk deployment.
Email	Maddy + Roundcube	Local and fleet mail only; no internet email.

6. The WAX Container Format

Frozen at v0.9 and implemented. A .wax file is a fixed header, a blob body, and a SQLite index footer, with the header committed last. That layout is what permits both writing and reading to stream.

6.1 Header — 128 bytes, little-endian

Offset	Field	Size	Notes
0–3	magic	4B	ASCII "WAX1". Readers reject anything else.
4	version_major	1B	A reader must refuse an unknown major version rather than guess.
5	version_minor	1B	Additive only. A reader ignores unknown minor-version index columns.
6–7	flags	2B	Bits 0–3: has_search_index, has_delta_base, is_multi_volume, is_signed.
8–23	archive_uuid	16B	Stable pack identity across versions. What a delta update matches against.
24–31	created_at	8B	Unix epoch seconds. Pack creation time, distinct from each segment's own write time.
32–39	index_offset	8B	Offset to the newest index segment. Rewritten on every append.
40–47	index_length	8B	Length of that segment.
48–55	blob_section_length	8B	Sum of every segment's blob region, for a fast integrity pre-check.
56–63	search_index_offset	8B	Offset to the v2 stand-off search segment. 0 = absent, which is all of v0.9.
64–71	search_index_length	8B	Length of that segment.
72–127	reserved	56B	Zero-filled. Readers ignore it entirely.

6.2 Index footer — SQLite, one database per segment

- **entries:** path TEXT PRIMARY KEY, title TEXT, offset, length, uncompressed_length, mime TEXT, compression TEXT, sha256 BLOB, volume_id INTEGER DEFAULT 0, redirect_to TEXT DEFAULT NULL. Declared WITHOUT ROWID: measured at 1.29 index page reads per lookup versus 2.03, with a path-ordered walk falling from 222,617 random reads to 11,563 sequential, and the index shrinking roughly 35%.
- **manifest:** key TEXT PRIMARY KEY, value TEXT. Lives only in the base segment and is immutable across appends — a manifest change is a new pack version, not an append.

- **segment_meta:** Per-segment key/value carrying format, segment index, blob region offset and length, created_at, and — on every non-base segment — back-links to the previous segment. This is how a reader walks the chain from the header's single pointer.
- **search_index:** Optional FTS5 virtual table at v0.9. Ownership of its schema and tokenizer contract is an open decision (§11).
- **signatures:** Defined but unwritten through v0.9/v1.x, which sign via an external sidecar. Activated only when TUF multi-key signing lands.

6.3 Semantics

- **Segment chain:** The header points at the newest segment; each non-base segment back-links to its predecessor. A reader walks backward to the base then merges forward, last-segment-wins per path. Primary keys are enforced per segment, not globally.
- **Append-commit protocol:** Append new blob bytes and a new index segment, fsync, then overwrite the fixed 128-byte header in place as the last atomic step. Only that final small fixed-size write touches existing bytes.
- **Redirects:** redirect_to holds the target's path in canonical form and must resolve to a non-redirect row. Chains are flattened to one hop at build time; cycles and dangling termini are dropped with a counted warning. A reader never follows more than one indirection.
- **Hashing:** sha256 is over the entry's uncompressed content, so it is stable across recompression. Verified by default on read, with an opt-out for byte-range serving.
- **Compression:** Permitted values are none and zstd. Any other value is a hard per-entry read error, never a silent skip.
- **Signing:** A digest over header bytes and all segment bytes in chain order, signed with minisign in prehashed mode as a detached <archive>.minisig sidecar, re-signed on every append, with archive_uuid bound into the trusted comment. Rollback replay is a documented residual risk, mitigated device-side by tracking the highest version seen and fully addressed at v2 by TUF.

7. Normative Cross-Component Values

These values are consumed by more than one component. They are authoritative here. A component restating one of them locally is descriptive, not binding; any change happens in one place.

7.1 Roles

- **Closed set:** admin, teacher, student, guest — lowercase ASCII literals. A none sentinel is returned when no profile matches and is never stored.
- **Default deny:** Any privileged operation not explicitly listed in the permission table is admin-only.
- **Passwordless roles:** A password hash is nullable for student and guest, never for admin or teacher.

7.2 Identity and derived accounts

- **Profile to account derivation:** localpart = "p" + the first 12 hex characters of the profile id, lowercase. Satisfies the strictest target grammar (Matrix), is stable across renames, and encodes no display name. Each service adapter also persists its own profile-to-account mapping.
- **archive_uuid text form:** Canonical lowercase hyphenated (8-4-4-4-12). Bare 32-hex is accepted on input and normalized on write; never emitted.

7.3 Progress event schema

Field	Type	Value
event_id	TEXT	Source-supplied, opaque. UNIQUE with event_source; writes are insert-or-ignore so a repeated sync is idempotent.

profile_id	TEXT	Derived by the receiver from the sending origin — never trusted from a caller-supplied field.
event_source	TEXT	Closed: wax_pack, kolibri.
pack_id	TEXT	wax_pack: the archive_uuid. kolibri: kolibri:<channel_id>:<content_node_id>.
event_type	TEXT	Closed: pack_opened, pack_closed, entry_viewed, unit_completed, quiz_scored. Unknown values stored verbatim, ignored by rollups.
timestamp	INTEGER	Unix epoch milliseconds, UTC.
seq	INTEGER	Monotonic per profile and source. Orders events when the clock is unset or steps — real on hardware with no RTC.
payload	TEXT	UTF-8 JSON object, 4 KiB maximum. Oversized rows rejected at write.

- **Guest emits nothing:** Any progress insert whose profile resolves to the guest role is rejected, so "nothing about a guest session is remembered" holds regardless of the emitter.

7.4 State vocabularies

- **Service health:** healthy, starting, unhealthy, restarting, failed, stopped. failed is terminal — automatic restart abandoned, not in progress.
- **Install and queue:** queued, fetching, verifying, installing, installed, update_available, failed_signature, failed_rollback, failed_transport, cancelled, removing. Verification failures are terminal and need admin action; transport failures retry with backoff.

7.5 Origins and hostnames

Pack content and administrative surfaces are on deliberately separate DNS zones. They shared one zone in an earlier draft, and because .lan is not on the Public Suffix List, the forward-auth session cookie scoped to that parent was readable and settable by every untrusted pack origin.

Surface	Origin
Shell UI	http://127.0.0.1:40999 – reserved, outside the pack port range
Pack content, pre-proxy	http://127.0.0.1:PORT, PORT = 41000 + (pack_index × 32) + profile_slot
Pack content, post-proxy	https://pack-<pack_index>-slot-<profile_slot>.deltos-packs.lan
Admin and service surfaces	https://<service>.deltos.lan

- **Ranges:** pack_index is 0–511, allocated at install and never reused. profile_slot is 0–31, assigned at profile creation and stable for that profile's life. The resulting port range 41000–57383 is reserved; no other service may bind inside it. A device therefore supports 32 concurrent profiles, and creation beyond that fails with a stated error rather than recycling a live slot.
- **Certificates:** Two separate wildcard certificates, *.deltos.lan and *.deltos-packs.lan, so the certificate fronting admin surfaces is not also the one fronting untrusted pack content.

7.6 Filesystem paths

Path	Holds
/var/lib/deltos/capability.json	The box's own capability record: profile, tier, measured RAM/storage/arch, radio capability, RTC presence.
/var/lib/deltos/b4/	On-device catalog database.
/var/lib/deltos/packs/	Installed .wax archives and signature sidecars.
/var/lib/deltos/packs-data/<archive_uuid>/	Per-pack writable storage. Created at install, removed at uninstall.

/var/lib/deltos/identity/	Profiles and progress events.
/var/lib/deltos/health/	Health snapshots.
/var/lib/deltos/<service>-adapter/	Each application adapter's profile-to-account mapping.

- **Rollback safety:** State under this tree carries a schema version its owner refuses to open if newer than it understands. /var is shared across OS deployments and never rolled back, so a rollback otherwise hands old code data that new code already migrated.

7.7 Formats and vocabularies

- **Language codes:** BCP-47, validated against the IANA subtag registry rather than merely checked for well-formedness — "english" and "zz" are both well-formed and both wrong, and a mistyped language degrades search silently.
- **Pack versions:** CalVer YYYY.MM.N: four-digit year, zero-padded month 01–12, and N an integer from 1 upward that increases within the month — a release counter for hand-built packs, the source date's day for converted ones.
- **License allowlist:** Literal SPDX identifiers, matched case-insensitively and normalized on write: CC0-1.0, CC-BY-4.0, CC-BY-SA-3.0, CC-BY-SA-4.0, GFDL-1.3-or-later, MIT, Apache-2.0, GPL-2.0-only, GPL-3.0-only, GPL-3.0-or-later.
- **Licensing outcomes — four, not three:** Blank with no operator value is a hard build failure. An allowlisted identifier builds clean. Free text or a recognized non-allowlisted identifier builds with a review-required outcome. An absent source license requires an operator-supplied value and always carries review-required — absence means unknown, not unlicensed, and no real ZIM including Wikipedia states a license at all.
- **MQTT topics:** deltos/v1/<device_class>/<device_id>/<metric>, device_class closed to battery, solar, grid, env, contact, aircraft, vessel; device_id matching [a-z0-9-]+ and stable per physical device; payload {"value", "unit", "ts"} retained at QoS 1. Commands use the reserved deltos/v1/actions/ prefix.

7.8 The build report

Every successful build writes <archive-filename>.build-report.json beside the archive, by default rather than on request — intake cannot depend on a report a builder may not have been asked to emit, and an absent report is a failed intake rather than an assumed-clean pack.

```
{ "report_version": 1, "archive_uuid": "...", "archive_filename": "...", "built_at": <epoch ms>,
  "builder_version": "...", "entry_count": N, "redirect_count": N, "skipped_count": N, "license":
  "CC-BY-SA-4.0", "license_review_required": false, "signed": true, "warnings": [ { "code":
  "unsupported_mimetype", "count": 1204 } ] }
```

- **Warning codes — closed vocabulary:** unsupported_mimetype, redirect_cycle, redirect_dangling, invalid_path, reserved_prefix_collision, license_operator_supplied, attribution_operator_supplied, icon_generated. Intake branches on these, so an invented code is a silently misrouted pack.
- **Not a trust artifact:** It sits outside the signature and is the builder's claim about its own run — useful for routing and diagnostics, never evidence about a pack's contents.

8. Content Conversion and Path Rules

These rules govern how external content becomes a WAX pack. They are stated here because they were the most expensive rules in the project to get right — several were wrong in the design documents and were only corrected by running a real Wikipedia archive through them — and because the crawler pipeline will need the identical path rules when it lands.

8.1 Path canonicalization

- **Article namespace flattens to the root:** The namespace prefix is stripped and an .html suffix applied: A/Photosynthesis becomes Photosynthesis.html.

- **Case is preserved, never folded:** This is the rule most likely to be got wrong and the most destructive to get wrong. A real Wikipedia archive carries 13th_Amendment, 13th_amendment and 13th_ammendment as three distinct entries, and 4H_disease alongside 4h_disease. Lowercasing collides them irreversibly and silently. Path comparison is exact and case-sensitive everywhere; normalization is a build-time obligation and never a reader behavior.
- **Non-article namespaces map to reserved prefixes:** `_assets/` for images and media, `_meta/` for metadata entries.
- **Classification is by mimetype, not namespace:** On any current-format ZIM the `C/` namespace holds articles, images, CSS and video together — `C/Michael_Jackson` beside `C/_assets/...` and `C/_mw/...`. An entry is an article if and only if its mimetype is `text/html`; everything else routes to `_assets/`. The same mimetype-first rule is applied to the legacy namespace scheme so one rule covers both.
- **In-document references are rewritten at conversion time:** Not resolved by the serving layer. Rewritten hrefs are root-relative, which is correct at any document depth because each pack is the root of its own origin.
- **Reserved prefixes can collide with real source paths:** A source entry whose own path begins `_assets/` or `_meta/` would shadow a converted asset. Such entries are dropped with a `reserved_prefix_collision` warning rather than silently overwriting.
- **Paths that cannot be canonicalized are dropped:** Real archives contain entries whose canonical path is invalid — a doubled separator, for instance. These are dropped with an `invalid_path` warning rather than written.

8.2 Redirects

- **Representation:** `redirect_to` holds the target's canonical path and must resolve to a non-redirect row. Source formats storing a target index rather than a path must translate.
- **Flattening:** Chains flatten to a single hop at build time, so no reader ever follows more than one indirection and no reader needs loop detection.
- **Pathological chains:** Cycles and chains whose terminus does not exist are dropped with counted warnings, never emitted.

8.3 Manifest derivation from source metadata

Manifest field	Derivation
name	Source Title metadata.
icon	The 48×48 illustration entry, else any other illustration entry, else the legacy favicon entry, else a generated placeholder raising <code>icon_generated</code> . Never omitted — it is a required field.
version	Source Date as CalVer. A YYYY-MM-DD source date reformats directly, with the day as the counter.
attribution	Creator, else Publisher. Where the source carries neither, an operator flag supplies it — never an empty string, which the schema rejects.
entry_point	The source's main-page pointer.
languages	Source Language metadata mapped from ISO 639-3 to BCP-47. Omitted where no mapping exists rather than guessed.
category, min_hw_tier	No source analog exists. Required operator flags; conversion fails if absent rather than defaulting.
license	Source license where stated. Where absent — which is every real archive examined, Wikipedia included — a required operator flag supplies it and the pack always carries review-required.

8.4 Content handling

- **Unsupported media:** Entries whose type the current converter does not carry are skipped, counted in the build report, and their references left intact rather than rewritten. Real Wikipedia archives carry pronunciation audio, so this is the common case rather than an edge case.

- **Placeholder titles:** Some producers write the literal string "null" as the title on every non-article entry. A title of exactly null on a non-HTML entry is treated as absent; the check is deliberately narrow so a genuine article titled "Null" survives.

9. The Existing Codebase

The implementation to date lives in one repository. A system picking this up should clone it and read the design documents committed alongside the code before changing anything.

Item	Value
Repository	github.com/TheDataStar/wax-format
Design documents	docs/ — thirteen markdown files. Read docs/INDEX.md first: it gives the reading order and states that the shared-values document overrides any component document where they disagree.
Format specification	SPEC.md at the repository root. Amended only under the precedence rule: it owns bytes inside the container, while the shared-values document owns how a shared value is rendered as text anywhere it is written.
Crates	wax-core (reader, writer, streaming, and a purpose-built read-only SQLite driver), wax-builder (library plus CLI), zim2wax (converter), fuzz/ (three libFuzzer targets).
CLI surfaces	wax-builder: build, append, inspect, verify, ls, read. zim2wax: conversion with the operator flags in §8.3.
Branches	main carries the documents. Implementation branches chain: track-a/spec-and-conformance, then track-a/wax-builder, then track-a/streaming-writer, then track-a/streaming-reader, which holds the most recent work.

9.1 Toolchain and build

- **Prerequisites:** Rust stable plus a C toolchain, required by the bundled SQLite and zstd. Nightly plus cargo-fuzz for the fuzz targets. A minisign binary for the signing tests — a password-less key, since unattended builds cannot answer a passphrase prompt.
- **Tests:** cargo test --workspace runs the suite. In CI, set the environment flag that makes a missing minisign a hard failure rather than a skip — the signing tests once reported green while silently skipping, and that flag exists so it cannot recur.
- **Reproducing the measurements:** The readbench example in wax-core reproduces every figure in §10, including index page-read counts, which are the numbers that transfer to SD-card storage.
- **Determinism:** Two builds of the same source tree are byte-identical given a pinned archive identifier and pinned timestamps. Both must be pinned independently: the pack's creation time and each segment's own write time are different clocks.

9.2 Known risk in the current code

- **The storage driver is untested outside Windows:** The read-only SQLite driver underpinning every pack open has so far been exercised only on Windows, while the product targets Linux on ARM. It delegates to the platform default driver and the Unix path is the same delegation, but the first test run on Linux is the real check and has not happened. This is the highest-risk untested surface in the codebase.

10. Implementation State

Everything below is committed, tested and reproducible. Track A's foundation is complete through its earliest phase and, unusually for this stage, measured against production-scale content rather than fixtures.

- **BUILT** A3 SPEC.md — the versioned byte-layout specification, frozen.

- **BUILT** A4 Conformance suite and fuzzers — 205 tests plus three libFuzzer targets over the header parser, index loader and segment merge. Millions of iterations, no crashes or hangs.
- **BUILT** A1 wax-core — reader and writer, both streaming. A purpose-built read-only SQLite driver windows the index directly out of the archive, so nothing is copied to open a pack.
- **BUILT** A2 wax-builder — directory tree to signed pack, full manifest schema enforcement, append support, build report.
- **BUILT** A7 signing, build-time half — detached minisign over a digest covering header and index together.
- **BUILT** B1 zim2wax — ZIM to WAX conversion, working against real Wikipedia rather than fixtures.
- **BUILT** B3 manifest schema and B6 licensing validator — enforced at build time with the four licensing outcomes.

Measured results

Three real Wikipedia archives spanning roughly 100× in size. Release build, heap sampled at 20 ms, confirmed under a kernel-enforced memory cap.

Archive	Size	Entries	Convert	Convert heap	Open cost
en_chemistry_mini	24 MB	58,257	4.6 s	23.5 MB	1.9 MB
en_100	318 MB	9,272	0.9 s	20.5 MB	1.9 MB
en_history_maxi	2,267 MB	382,608	63 s	55.9 MB	1.9 MB

- **Conversion cannot exhaust a small box:** The 2.27 GB conversion completes inside a hard 128 MB commit limit with the kernel reporting 57.1 MB peak — a process that could not have exceeded the cap, not an estimate from a memory graph.
- **Opening a large pack is free:** Before this work, opening the 382,608-entry pack cost 332 MB — roughly 870 bytes per entry, extrapolating to 5–6 GB at full Wikipedia, meaning the pack a Kiosk box exists to serve was the pack it could not open. It now costs 1.9 MB regardless of size, and the whole pack verifies under a 64 MB cap.
- **Reads stay cheap on slow storage:** 1.3 index page reads per lookup and 1.6 per full read — the figure that transfers to SD cards at 1–2 ms per random read. A bounded lookup cache plus SQLite's page cache absorb a hot set to roughly a quarter of one page read per operation.
- **Conversion is a publishing-side operation:** Extrapolated scratch space for a full Wikipedia conversion is roughly 1.7 GB on disk, against a Kiosk storage floor of 16 GB and a 100 GB output. Conversion runs where the catalog publishes, on Community Hub-class hardware or better. A device consumes packs; it does not build them.

11. Open Decisions — Blocking

These must be answered before the work that depends on them. None is a coding task; each is a decision. They must not be resolved silently by an implementer.

- **BLOCKED** Embedded search index ownership. Converted packs are browse-only today — every link and image works, nothing is searchable — because the index's schema and tokenizer contract sit unresolved between the format track and the search track. It is the oldest open decision and the only one now blocking a shipped capability rather than a future one. Every pack built before it is settled inherits the gap.
- **BLOCKED** The shell's privilege boundary. The shell splits into an unprivileged web app and a privileged helper, and the interface between them — wire format, operations, argument types, error taxonomy, who may call what — is named but never specified. It is the single largest gap in the design set and a design session rather than a value to pin. Nothing in the shell track can be built correctly without it. A related unresolved fork: browser JavaScript cannot open a Unix socket, so whether the bridge is a loopback WebSocket or a helper process determines the entire authentication model.
- **BLOCKED** Per-service resource floors for the application catalog. Ten third-party services are scheduled to run concurrently on one box and not one has a stated memory or disk floor. The consolidated hardware budget is

waiting on exactly these numbers, and the tracks that own the components are the only ones that can supply them.

- **BLOCKED** Service sessions versus profile switching. Application services are long-running and hold their own cookie sessions at fixed hostnames, so switching profiles in the shell leaves the previous profile's session authenticated — the cross-profile leak the identity design exists to prevent. No mechanism currently binds a service session to the active profile.
- **BLOCKED** The passage unit for search and citation. What gets embedded, retrieved and cited is undefined: no chunk size, overlap, boundary rule or chunk identifier. A citation cannot locate a passage inside a large article, and two implementations would produce incompatible vector data for the same pack.
- **BLOCKED** Cross-source search ranking. Scores from separately-built indexes are not comparable — FTS5 relevance scores are corpus-relative — so a normalization must be chosen. Reciprocal Rank Fusion is the obvious candidate but the choice belongs to the search track.
- **BLOCKED** Update failure detection. The OS track specifies what a rollback restores but never what triggers one: what counts as a healthy boot, what detects failure, how many failures roll back. An atomic update system without a trigger does not roll back — an unconfigured deployment simply keeps booting the broken tree.
- **BLOCKED** Design system accessibility values. Five colour pairings fail the design system's own stated WCAG AA guarantee, including the primary action colour, which cannot reach 4.5:1 against any defined foreground in either polarity. There is also no focus-indicator token, no spacing scale, and no touch-target minimum, in a system built for touch on shared devices used by young learners. These values need replacing before any interface is built from them.

12. Known Defects

A pre-implementation audit read all nine design documents for implementability rather than consistency — not "do these agree?" but "could an engineer build this without inventing something?" It found roughly 140 defects, catalogued by document and severity. They matter to an implementing system because they mark where a document is not yet safe to build from.

The seven defect classes

Class	Defect	Example from this project
A	Undefined value domain	A flags field naming four flags and assigning no bit numbers; a compression column with no permitted values.
B	Self-referential or unmaintainable value	A manifest field recording the archive's own size, from inside that archive, in a table declared immutable.
C	Dangling referent	A dependency field carrying "pack ids" in the same table whose next note deletes the id field.
D	Mechanism named but not mechanized	"The reader merges the segment chain" — with no way to find segments 2..N from a pointer to one.
E	Unowned or mis-sequenced dependency	Signing tagged for a later phase while three other tracks assumed it live earlier.
F	Internal contradiction	A pointer described as fixed at the base segment and as rewritten on every append.
G	Unsatisfiable requirement	Requiring both a random identifier per build and byte-identical rebuilds.

The six systemic causes

- Shared vocabularies had no owning document — each was defined loosely wherever first needed and restated differently by every consumer. This produced most of classes A and C and is the reason §7 of this document exists.

- Deployment profiles and hardware tiers were repeatedly conflated, because no mapping or ordering between the two axes had been published.
- Hardware budgets were circular — every track deferred to a consolidated budget that was waiting on numbers from those same tracks.
- Every trust boundary was asserted as a conclusion rather than specified as a mechanism.
- Amendments were recorded where they were decided rather than in the document that owned the field, so agreed changes silently failed to land.
- Example values did load-bearing work — an "e.g." where a normative value was required, in places where the example was the specification.

Defects found in real implementation are the most valuable and the least predictable. Contact with an actual Wikipedia archive exposed a path-canonicalization example that would have silently collided distinct articles, a licensing rule that forbade converting the flagship content, and a memory bound that held per byte and failed per entry. No amount of document review would have surfaced any of them.

13. Phase Roadmap

Phase	Theme	State
P0	Format frozen, conformance suite standing, first content converted, first-boot security, storage layout.	Complete for the format track, with B1 pulled forward from P1 and streaming on both sides added that was not originally scoped.
P1	Streaming server, second-language reference reader, on-device catalog, local DNS and NTP, atomic updates, admin console, onboarding.	Next. The streaming server is unblocked; the reference reader validates the now-frozen spec.
P2	Embedded search index, signing enforced by default, orchestration and ingress, media server, install pipeline, crawler, publishing service.	Gated on the search-index ownership decision.
P3	Multi-volume archives, LMS, cloud storage, wiki, chat, progress dashboard, offline maps and geocoding, local LLM.	Design settled, not started.
P4	Delta engine, SSO, password vault, office suite, browser IDE, email, VoIP, automation, metrics, mesh, IoT.	Design settled, not started.
P5	Semantic search at scale, federated search, SDR, authoring studio, format frozen as v2.	Scoped.

Immediate next work items

- **NEXT** Validate the storage layer on Linux and ARM. The read-only SQLite driver underpinning every pack open has so far been exercised only on Windows, while the entire product targets Linux on ARM. Highest-risk untested surface in the codebase.
- **NEXT** Settle the embedded search index ownership question (§11), then implement it. This unblocks search for every pack already built.
- **NEXT** A5 wax-serve — the HTTP server the shell needs before it can open a pack at all. Unblocked by the streaming reader.
- **NEXT** F8 governance — publish the specification, contribution guidelines, license posture and roadmap. Cheap, unblocked, and overdue for a project intended to be adopted rather than merely built.
- **NEXT** Stand up physical test hardware: one Pi 4 or 5 and one x86 rig running the current prototype end to end. Power-loss atomicity of the header write, real SD-card latency and Wi-Fi AP behavior cannot be settled anywhere else.
- **NEXT** A10 second-language reference reader — proves the specification is portable rather than merely implemented once, and must implement the segment-merge behavior, not just parse a single segment.

14. How This Work Has Been Run

Two roles, deliberately separated. Design documents are written and maintained as one body of work; implementation proceeds against them in scoped, titled prompts, one component at a time. The documents live in the repository alongside the code, so an implementer reads the specification rather than a paraphrase of it.

- **One document owns every shared value:** Section 7 of this handoff is that document in miniature. It exists because the audit found most defects traced to one cause: values defined loosely wherever first needed and restated differently by each consumer. Where it and a component document disagree, it wins, and a change happens in one place.
- **The rule that makes it work:** An implementer who needs a value that the shared document should own, and does not find it, files a defect against that document rather than choosing quietly. This is the single most important instruction in this handoff.

The measurable effect: recent implementation passes produced defects reported against the documents and no invented values in the code. Five passes halted outright rather than proceed on a guess, and every halt was correct — one had found a specification that did not exist, one a stale document, one a file that failed a checksum, one a missing precondition, and one a set of answers that had changed since the brief was written.

- **What an implementing system should carry forward:** Treat an undefined value as a stop condition, not a judgment call. Verify a document's identity before building from it. Prefer measuring to estimating — every significant result in §10 came from running real content through real code under a real memory cap, and several of the most valuable defects were invisible until that happened.